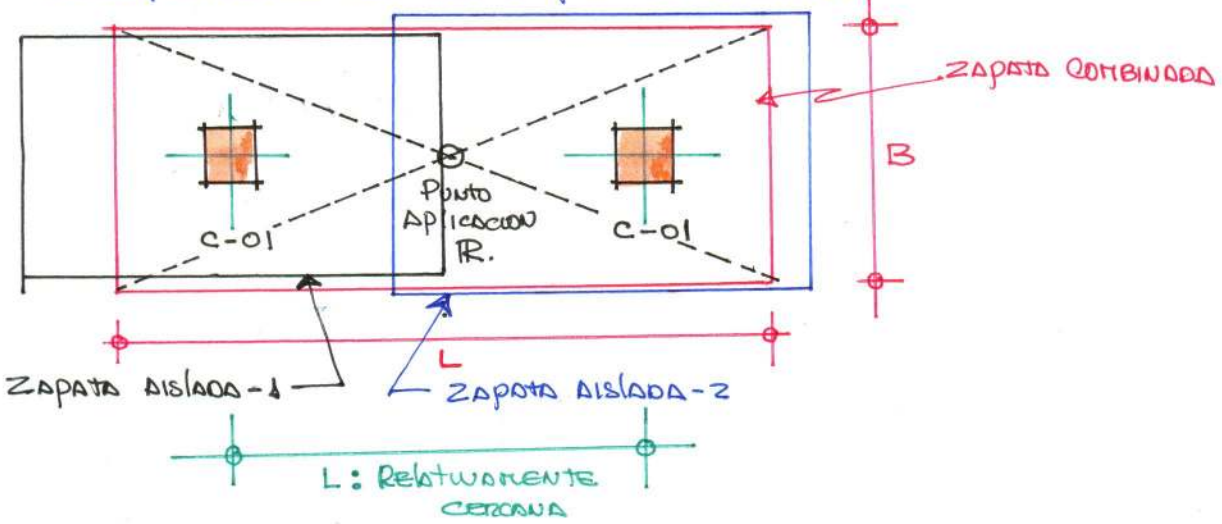
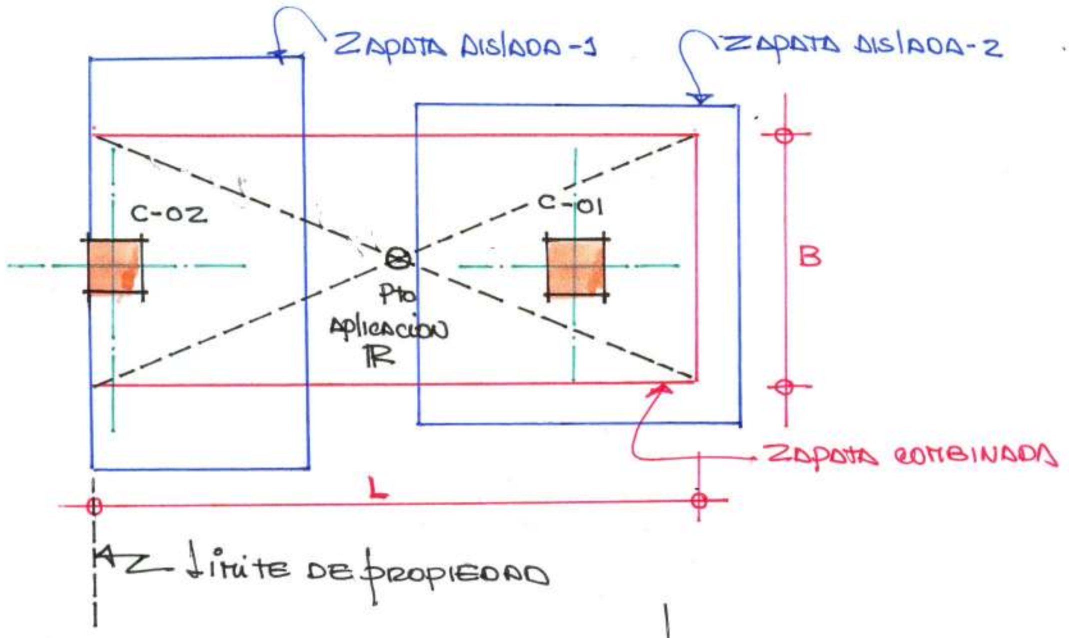
ZAPATAS COMBINADAS

UNA CIMENTACIÓN COMBINADA RECIBE LAS CARGAS PROVENIENTES DE LA SUPER ESTRUCTURA ATRAVEZ DE 2 O MAS COLUMNAS, SE USA PRINCIPALMENTE EN LOS SIGUIENTES CASOS:

a) CUANDO DOS COLUMNAS ESTAN RELATIVAMENTE CERCANA ENTRE SI, DE MODO QUE SE UNIRAN LAS ZAPATAS AISLADAS ESTAS PODRAN TRANSFORMARSE EN UNA ZAPATA COMBINADA.



b) CUANDO UNA COLUMNA EXTERIOR ESTA EN EL LIMITE DE PROPIEDAD O MUY CERCANA DE EL, DE MODO QUE SE UNIRA UNA ZAPATA AISLADA ESTA RESULTARIA CON CARGA EXCENTRICA EXCESIVA.



EN AMBOS CASOS EL FACTOR ECONOMICO TAMBIEN ES DETERMINANTE

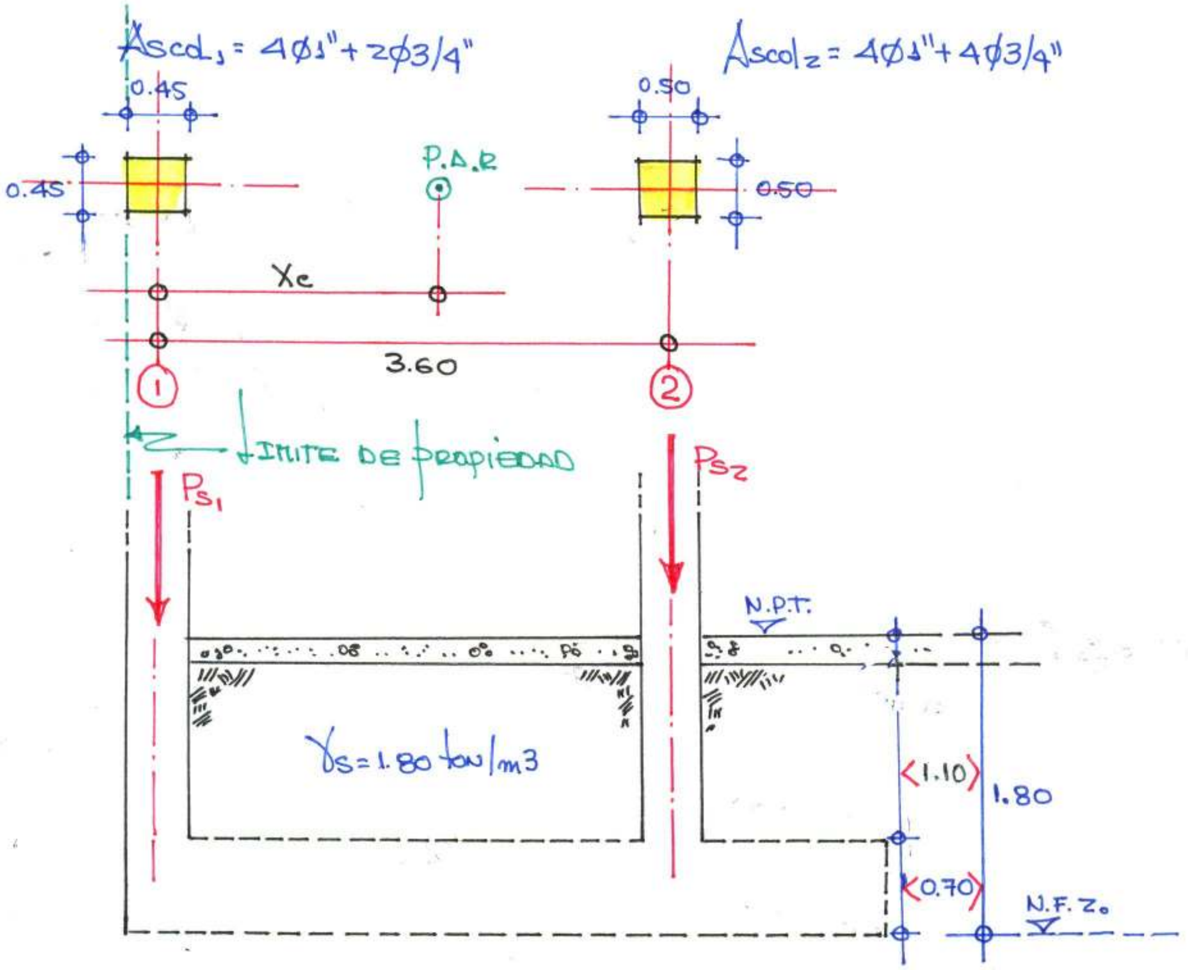
PROBLEMA Nº 01:

DISEÑAR LA ZAPATA COMBINADA DE LA FIG. MOSTRADA PARA LAS SOLICITACIONES INDICADAS:

$P_{CM_1} = 55$ ton $P_{CM_2} = 90$ ton $f'c = 280$ Kg/cm²

$P_{CV_1} = 45$ ton $P_{CV_2} = 70$ ton $fy = 4200$ Kg/cm²

$\sigma_t = 2.40$ Kg/cm² NF = -1.80 m.



DIMENSIONAMIENTO:

4 PERALTE: Acero columna C-02: 4Ø1" + 4Ø3/4" ⇒ Ø1" { db = 2.54 cm. ; Ab = 5.10 cm² }

Consideraciones Normativas

$ld = 0.08\, db\, fy/\sqrt{f'c}$ $ld = 51.003$ cm.

$ld = 0.004\, db\, fy$ $ld = 42.672$ cm. ⇒ $ld = 50$ cm.

$ld \geq 20$ cm. $ld = 20.00$ cm.

$1d = 50\,cm \Rightarrow d = 60\,cm, \quad \therefore h = 70\,cm.$

▲ Capacidad portante neta del terreno : (q_e)

$$q_e = \sigma_t - (\gamma_{c^\circ z} * h_z) - (\gamma_s * h_s)$$

$$= 24.0 - (2.40 * 0.70) - (1.80 * 1.10) \Rightarrow q_e = 20.34\ ton/m^2$$

$$q_e = 2.034\ kg/cm^2$$

▲ Solicitaciones

↳ Cargas de Servicio : $P_s = P_{cm_1} + P_{cv_1} \Rightarrow P_{s_1} = 55+45 \Rightarrow P_{s_1} = 100\ ton$

$P_{s_2} = P_{cm_2} + P_{cv_2} \Rightarrow P_{s_2} = 90+70 \Rightarrow P_{s_2} = 160\ ton$

$P_s = 260\ ton$

↳ Cargas ultimas

$$P_{u_1} = 1.40\,P_{cm_1} + 1.70\,P_{cv_1} \Rightarrow P_{u_1} = 1.40(55)+1.70(45) \Rightarrow P_{u_1} = 153.50\ ton$$

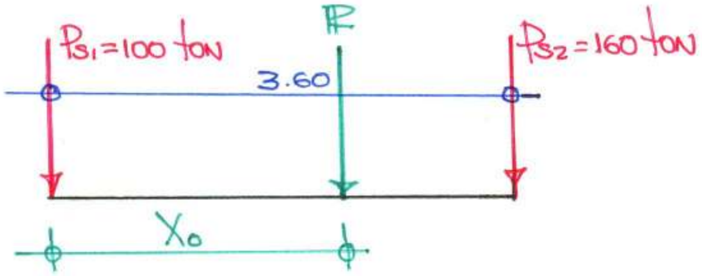
$$P_{u_2} = 1.40\,P_{cm_2} + 1.70\,P_{cv_2} \Rightarrow P_{u_2} = 1.40(90)+1.70(70) \Rightarrow P_{u_2} = 245.00\ ton$$

$$P_u = 398.50\ ton$$

▲ Area Requerida :

$$\boxed{A = \frac{P_s}{q_e}} \Rightarrow A = \frac{260.00}{20.34} \Rightarrow A = 12.783\ m^2$$

▲ Magnitud de la Resultante



$\Sigma F_V = 0$

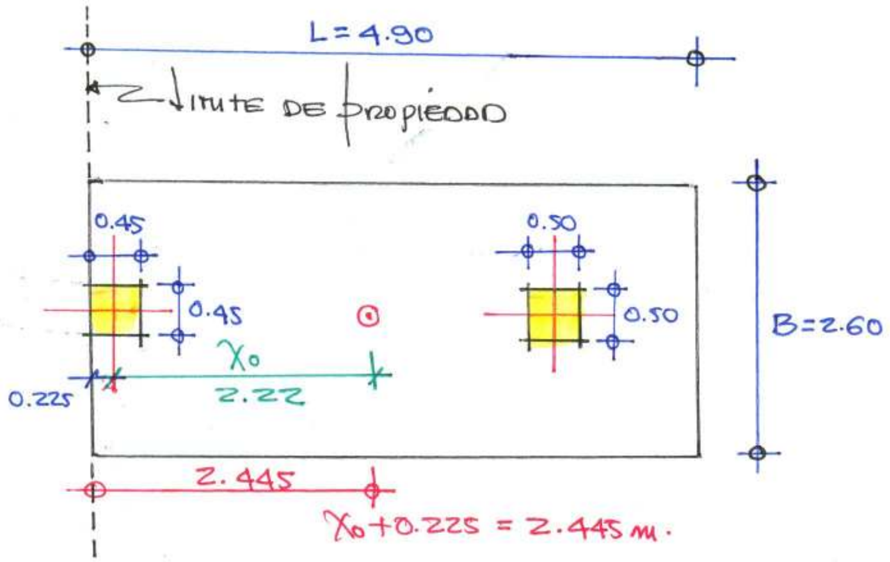
$-100 - 160 - R = 0$

$R = -260\ ton$

$\Sigma M_b = 0$ (+) (-)

$-R(X_o) - 160\,(3.60) = 0$

$X_o = 2.22$



LONGITUD DE LA ZAPATA $L = 2(2.445) \Rightarrow L = 4.89 \Rightarrow \underline{L = 4.90\ m}$

$\therefore \boxed{B = \frac{A}{L}} \Rightarrow B = \frac{12.783}{4.90} \Rightarrow B = 2.609\ m \longrightarrow \underline{B = 2.60\ m}$

$\underline{A = 12.74\ m^2}$

PRESIONES DE DISEÑO :

$P_{U1} = 153.50$ ton

$P_{U2} = 245.00$ ton

$P_U = 398.50$ ton

↳ CARGAS DISTRIBUIDAS :

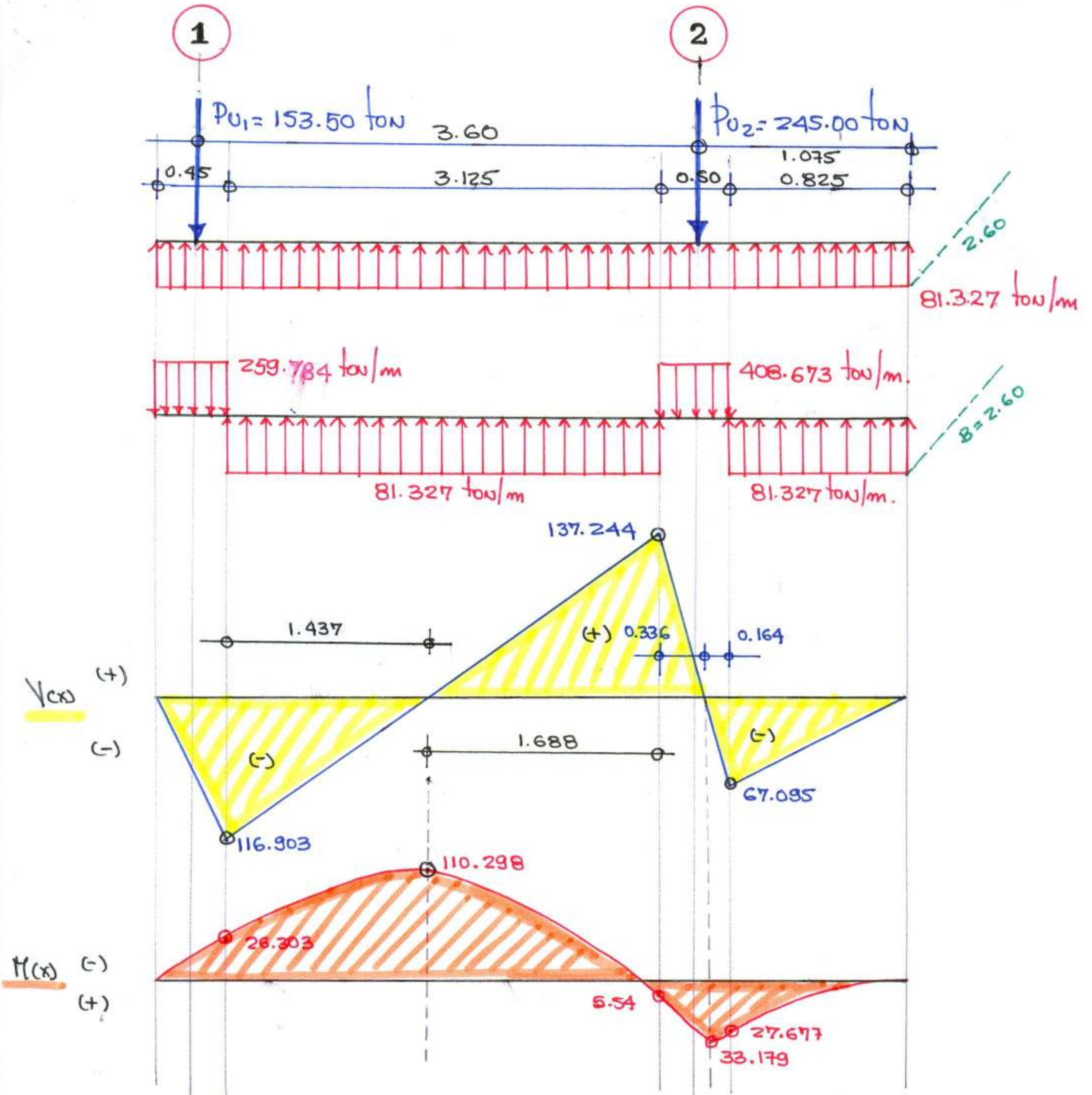
$A = 12.74\ m^2$, $L = 4.90\ m$, $B = 2.60\ m$.

$\boxed{q'_U = \frac{P_U}{A}} \Rightarrow q'_U = \frac{398.50}{12.74} \Rightarrow q'_U = 31.279$ ton/m²

$\therefore q_U = 31.279\ (B)$

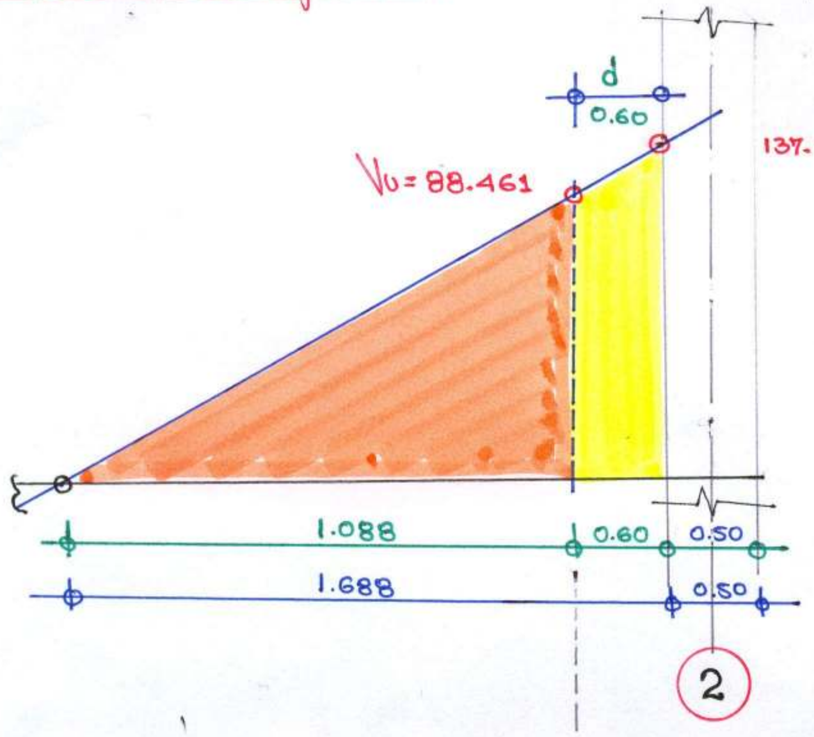
$q_U = 31.279\ (2.60)$

$\underline{q_U = 81.327\ \text{ton/m}}$



DIAGRAMAS ESFUERZO CORTANTE Y MOMENTO FLECTOR

Verificacion Corte-flexion: $B=2.60$, $L=4.90$, $A=12.74\ m^2$ $d=60$ cm.

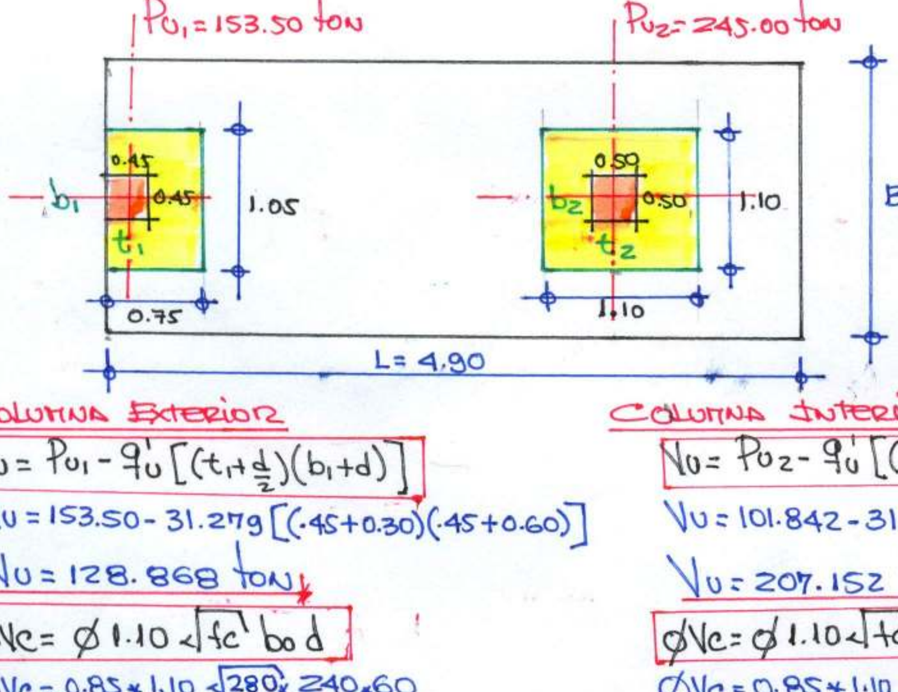


$\boxed{V_u \le \phi V_c}$

$V_u = 88.461$ ton $\quad \phi V_c = \phi\, 0.53 \sqrt{f'c}\, B\, d = 0.85 * 0.53 \sqrt{2} * 260 * 60 = 117.597$ ton

$V_u = 88.461$ ton $< \phi V_c = 117.597$ ton (cumple) conforme

Verificacion por punzonamiento



$\boxed{V_u \le \phi V_c}$

Columna Exterior

$\boxed{V_u = P_{u1} - q'_u\left[\left(t_1+\frac{d}{2}\right)(b_1+d)\right]}$

$V_u = 153.50 - 31.279[(.45+0.30)(.45+0.60)]$

$V_u = 128.868$ ton

$\boxed{\phi V_c = \phi\, 1.10 \sqrt{f'c}\, b_o d}$

$\phi V_c = 0.85 * 1.10 \sqrt{280} * 240 * 60$

$\phi V_c = 225.296$ ton

$V_u < \phi V_c$ (Conforme)

Columna Interior

$\boxed{V_u = P_{u2} - q'_u\left[(t_2+d)(b_2+d)\right]}$

$V_u = 101.842 - 31.279[(.50+.60)(.50+.60)]$

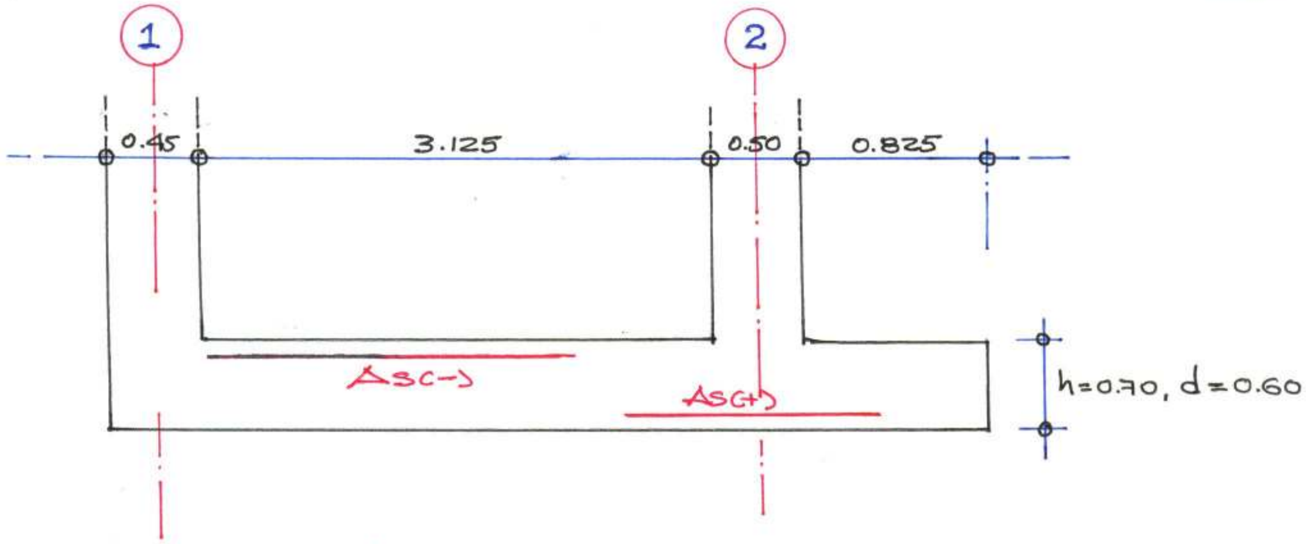
$V_u = 207.152$ ton

$\boxed{\phi V_c = \phi\, 1.10 \sqrt{f'c}\, b_o d}$

$\phi V_c = 0.85 * 1.10 \sqrt{280} * 440 * 60$

$\phi V_c = 413.042$ ton

$V_u < \phi V_c$ (Conforme)



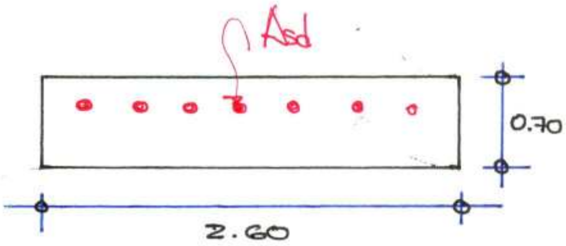
DISEÑO REFUERZO LONGITUDINAL:

① TRAMO: 1-2 ACERO NEGATIVO

DATOS DE DISEÑO:

$M_u = 110.298$ ton-m $f'c = 280$ kg/cm²

$b = 260$ cm, $d = 60$ cm. $f_y = 4200$ kg/cm²



$$\boxed{a = \frac{As\,f_y}{0.85\,f'c\,b}} \qquad \boxed{M_u = \phi\,As\,f_y\,(d - a/2)}$$

$\rho_{min} = \frac{0.70\sqrt{f'c}}{f_y}$ $\rho_{min} = 0.00279 \Rightarrow As_{min} = \rho_{min}\,b\,d \Rightarrow As_{min} = 43.506$ cm²

$\rho_b = 0.85\beta_1 \frac{f'c}{f_y}\left(\frac{6000}{6000+f_y}\right)$ $\rho_b = 0.02833 \Rightarrow As_b = \rho_b\,b\,d \Rightarrow As_b = 441.948$

$\rho_d = w\, f'c/f_y$ $\rho_d = 0.00321 \Rightarrow As_d = \rho_d\,b\,d \Rightarrow As_d = 50.054$

$\rho_{max} = 0.75\,\rho_b$ $\rho_{max} = 0.02125 \Rightarrow As_{max} = \rho_{max}\,b\,d \Rightarrow As_{max} = 331.50$

DISTRIBUCIÓN DE REFUERZO:

$\phi 3/4 = 2.84$ cm², $d_b = 0.01905$ cm.

⇒ ACERO MINIMO: $As_{min} = 43.506$ cm², $m = \frac{As_{min}}{As\phi} \Rightarrow m = \frac{43.506}{2.84} \Rightarrow m = 15$ UNIDADES   50.054

$\therefore S = \frac{2.60 - 0.15 - 0.01905 - 0.10}{14} \Rightarrow S = 0.166 \Rightarrow S = 0.17$ cm.   0.14

⇒ ACERO DE REFUERZO: $As_R = As_d - As_{min} \Rightarrow As_R = 6.548$ cm²

$m = \frac{As_R}{As\phi} \Rightarrow m = \frac{6.548}{2.84} \Rightarrow m = 2$ UNIDADES. $\Rightarrow S = 0.75$

$\therefore$ As_{min}: 15ϕ3/4 @ 0.17 , As_R : 2ϕ3/4 @ 0.75

② Apoyo 2: Acero positivo

$M_u = 33.179$ ton-m $\quad f'c = 280$ kg/cm²

Datos de Diseño: $b = 260$ cm, $d = 60$ cm, $\quad f'y = 4200$ kg/cm²

$$a = \frac{As\,fy}{0.85\,f'c\,b} \Rightarrow Mu = \phi\,As\,fy\,(d - a/2)$$

$$\rho_{min} = 0.70\sqrt{f'c}/fy \Rightarrow \rho_{MIN} = 0.00279 \quad As_{MIN} = \rho_{MIN}\,b\,d \Rightarrow As_{MIN} = 43.506\ cm^2$$

$$\rho_b = 0.85\beta\frac{f'c}{fy}\left(\frac{6000}{6000+fy}\right) \Rightarrow \rho_b = 0.02833 \quad As_b = \rho_b\,b\,d \Rightarrow As_b = 441.948\ cm^2$$

$$\rho_d = w\,f'c/fy \Rightarrow \rho_d = 0.00095 \quad As_d = \rho_d\,b\,d \Rightarrow As_d = 14.753\ cm^2$$

$$\rho_{max} = 0.75\,\rho_b \Rightarrow \rho_{max} = 0.02125 \quad As_{MAX} = \rho_{max}\,b\,d \Rightarrow As_{MAX} = 331.50\ cm^2$$

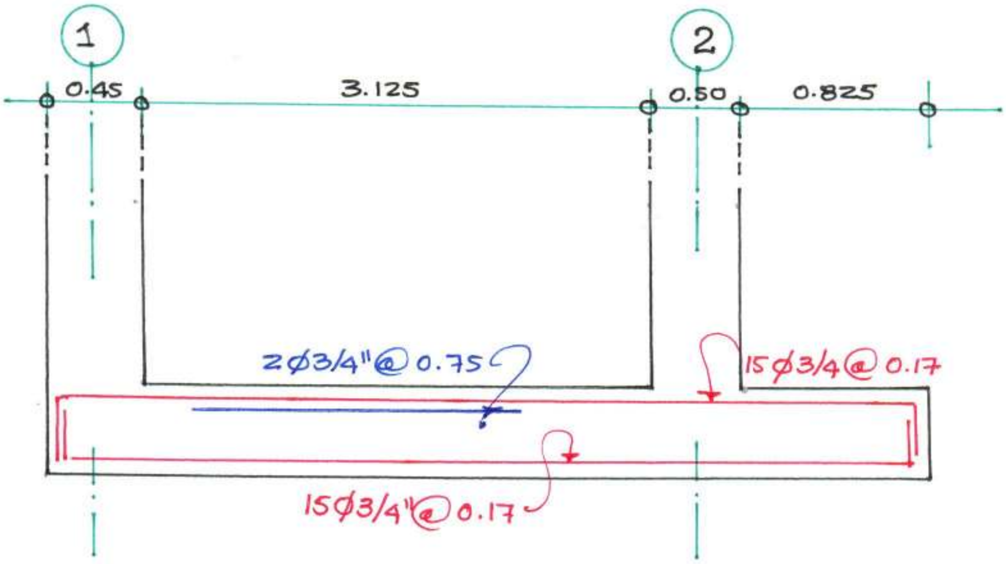
$\therefore As_d = As_{MIN} = 43.506\ cm^2$

Distribución de Refuerzo: $\phi 3/4'' = 2.84\ cm^2$, $d_b = 0.01905$ cm.

$$n = \frac{As_d}{A\phi} \Rightarrow n = \frac{43.506}{2.84} \Rightarrow n = 15.319 \Rightarrow n = 15 \text{ unidades.}$$

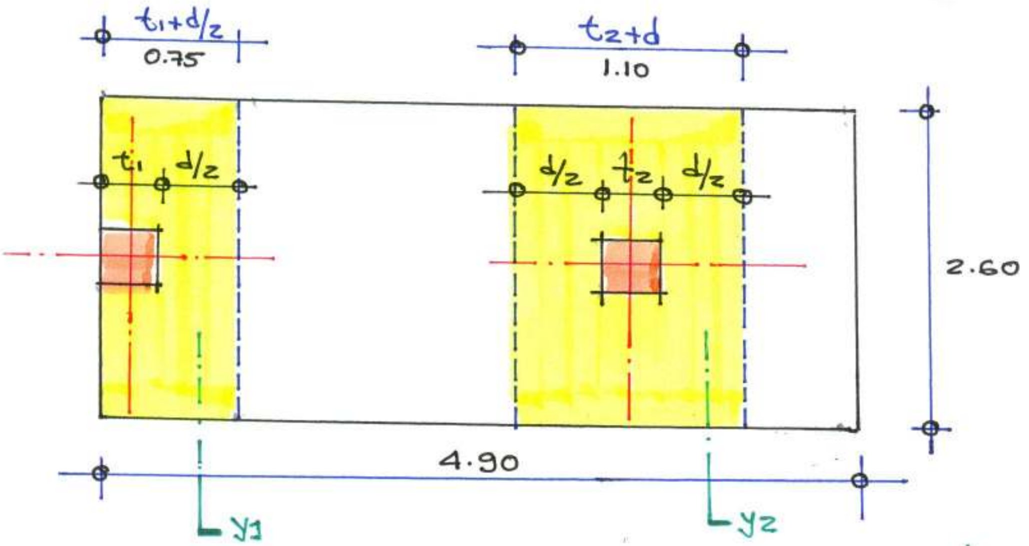
$$S = \frac{2.60 - 0.15 - 0.01905 - 0.10}{14} \Rightarrow S = 0.1665 \Rightarrow S = 0.17\ m.$$

$15\phi 3/4$ @ 0.17



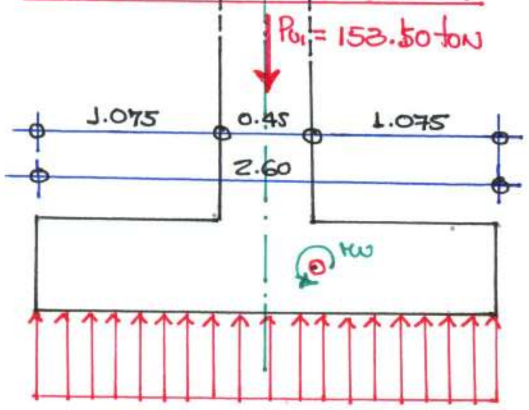
Detalle: Refuerzo Longitudinal

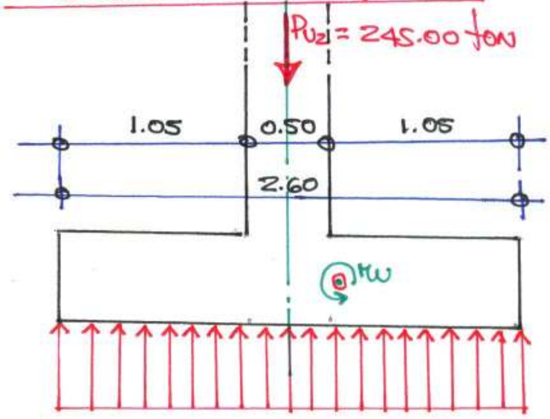
DISEÑO REFUERZO TRANSVERSAL



COLUMNA EXTERIOR (Y1-Y1)

COLUMNA INTERIOR (Y2-Y2)





$\frac{153.50}{2.60} = 59.038$ ton/m

$\frac{245}{2.60} = 94.231$ ton/m.

$\boxed{Mu = \frac{\omega L^2}{2}} \Rightarrow Mu = 34.113$ ton-m

$\boxed{Mu = \frac{\omega l^2}{2}} \Rightarrow Mu = 51.945$ ton-m

$b = 100$ cm, $d = 60$ cm, $f'c = 280$ kg/cm², $fy = 4200$ kg/cm²

$\rho_{min} = 0.00279$	$As_{min} = 16.733$ cm²	$\rho_{min} = 0.00279$	$As_{min} = 16.733$ cm²
$\rho_b = 0.02833$	$As_b = 169.98$ cm²	$\rho_b = 0.02833$	$As_b = 169.98$ cm²
$\rho_d = 0.00257$	$As_d = 15.39$ cm²	$\rho_d = 0.00396$	$As_d = 23.734$ cm²
$\rho_{max} = 0.02125$	$As_{max} = 127.50$ cm²	$\rho_{max} = 0.02125$	$As_{max} = 127.50$ cm²

$\Rightarrow Astd = Asd\,(0.75) \Rightarrow Astd = 11.54$ cm²

$n = 4.06$ Und, $n = 4$ Und.

$S = \frac{0.75 - 0.075 - 0.01905 - 0.10}{4} \Rightarrow S = 0.185$ m

$4 \phi 3/4 @ 0.185$ m.

$Astd = Asd\,(1.10) \Rightarrow Astd = 26.107$ cm²

$n = 9.19$ Und. $\Rightarrow n = 9$ Und.

$S = \frac{1.10 - 0.01905 - 0.10}{8} \Rightarrow S = 0.12$ m.

$9 \phi 3/4 @ 0.12$ m.

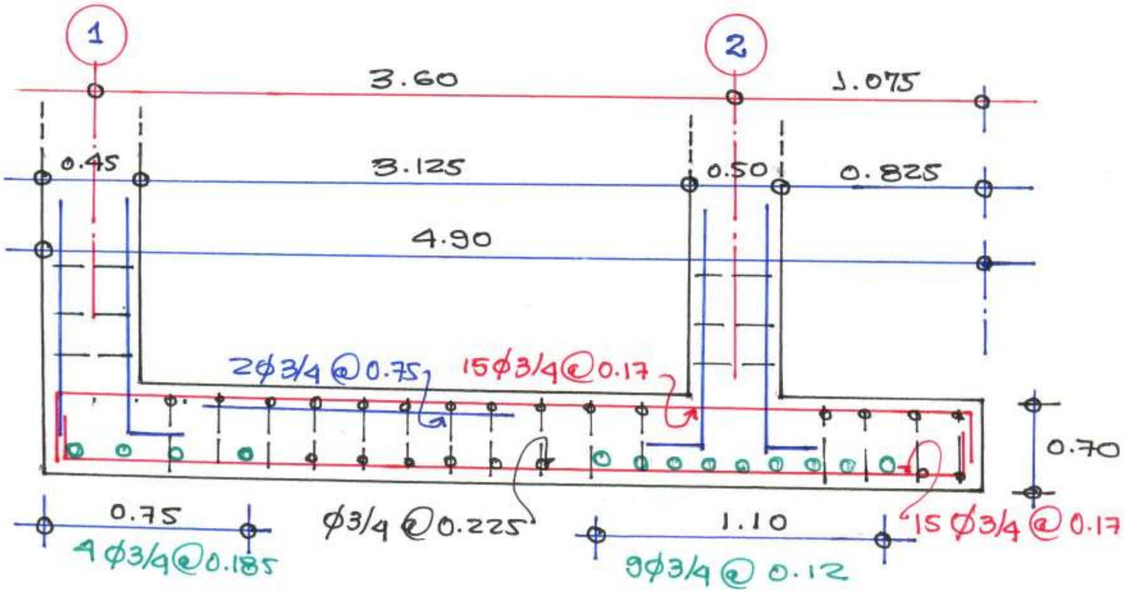
ACERO DE MONTAJE: $b = 100\ cm,\ h = 70\ cm,\ f'c = 280\ kg/cm^2,\ fy = 4200\ kg/cm^2$

$$\boxed{A_{SR} = 0.0018\ b\ h}$$

$A_{SR} = 0.0018 * 100 * 70 \Rightarrow A_{SR} = 12.60\ cm^2$

$$\boxed{S = \frac{b * A_{S\phi}}{A_{SR}}} \Rightarrow S = \frac{100 * 2.84}{12.60} \Rightarrow S = 22.50\ cm.$$

∴ A_{SR}: $\phi 3/4$ @ 0.225



DETALLE DE REFUERZO LONGITUDINAL TRANSVERSAL